swirl

# Software that thinks.

AI-native enterprise platform for hospitality.
POS, inventory, accounting, CRM — all working as one.

- Swirl Dine
- Swirl Works
- Swirl OS

# Built for the future of hospitality

Swirl Technologies is the holding company behind a suite of AI-native enterprise products that transform how businesses operate — embedding intelligence into every workflow, not bolting it on as an afterthought.

Swirl products don't just automate tasks. They eliminate the need to manage them at all. Every module is designed with a single principle: make the complex invisible.

### AI-First Philosophy

Intelligence embedded at the core, not layered on top. Every feature is designed to think, anticipate, and act autonomously.

### Unified Platform

One platform for front-of-house, back-of-house, and enterprise operations. No fragmented point solutions.

### Enterprise Scale

Multi-entity, multi-brand, multi-country. Architected from day one for chains operating hundreds of outlets.

THE CHALLENGE

# Enterprise F&B Is Running on Fragmented Tech

Multiple disconnected systems creating data silos, revenue leaks, and operational blind spots across every outlet.

### 5-10 Disconnected Vendors

POS, inventory, reservations, loyalty, payroll — all separate, all siloed.

**44%** of restaurant operators struggle with multi-platform integration daily

### No Single Source of Truth

Finance doesn't match ops. Reports take days to reconcile across outlets.

### Zero Real-Time Visibility

Managers have no live view of sales, wastage, or staff across branches.

### Manual Processes Everywhere

Purchase orders, payroll, banquet quotes — all done in spreadsheets.

### Revenue Leaking Daily

Aggregator fees untracked. Menu profitability unknown. Wastage uncontrolled.

# Three products. One platform.

FRONT OF HOUSE

## Swirl Dine

*From the first tap to the last bite.*

POS & Billing
Menu & QR
CRM & Loyalty
Reservations
KDS & Kitchen

BACK OF HOUSE

## Swirl Works

*Less waste. More margin. Zero chaos.*

Inventory
Procurement
Recipes & Costing
Production
Vendor Mgmt

PLATFORM LAYER

## Swirl OS

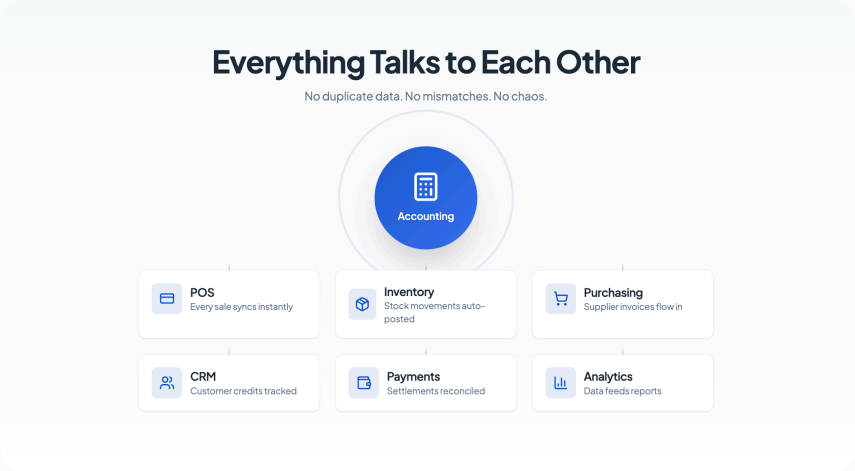
*Run everything. Manage nothing.*

Finance & Accounting
HR & Payroll
Multi-Entity
Fixed Assets

SWIRL PLATFORM — UNIFIED INTELLIGENCE

# Everything talks to everything.

One platform where POS, kitchen, inventory, accounting, CRM, and delivery all share the same data layer. No integrations to manage, no sync failures, no data silos — every module is natively connected.



# From the first tap to the last bite.

Every customer touchpoint — from online discovery to in-store dining — managed from a single intelligent platform. Swirl Dine unifies digital and physical guest experiences.

### Point of Sale

Touch POS for dine-in, takeaway, delivery. Multi-device, offline-capable, with split payments.

### Menu & QR Ordering

Digital menus, table-side QR ordering, multi-language. Customers order from their phone.

### CRM & Loyalty

Customer profiles, RFM segmentation, tiered loyalty, cross-brand redemption, automated campaigns.

### Kitchen Display

Multi-station KDS, automated routing, cook acknowledgement, prep timers, performance analytics.

### Table & Reservations

Visual floor layout, drag-and-drop tables, booking with WhatsApp/SMS reminders, waitlist management.

### Reports & Analytics

Sales, cashier, tax, voids. AI Command Center with daily insights. Live dashboard on mobile.

# Every guest touchpoint. Unified.

### Order Types

Dine-in, takeaway, delivery, online, kiosk. Each with configurable pricing and tax rules.

### Combos & Modifiers

Template combos, modifier groups, substitution surcharges, auto-detect from cart, happy hour scheduling.

### Bill Splitting

Equal, by item, by category, complimentary. Independent payment per sub-bill. Merge and transfer tables.

### Course Management

Sequential/parallel serving, auto-fire next course, prep buffers, course-level time tracking.

### Dynamic Pricing

Unlimited price contexts: order type, platform, time, day, area, customer segment. Fully rule-based.

### Queue & Tokens

QSR token system with display board, SMS/WhatsApp ready notification, daily auto-reset.

### Payments

Card terminals (Adyen, NI, SumUp), split payment, tips, gift cards, payment links, loyalty redemption.

### Aggregator Hub

Talabat, Noon, Careem, Deliveroo via UrbanPiper. Unified dashboard, auto-KOT, commission reconciliation.

### Allergen Safety

Ingredient-level tagging, customer profile alerts, POS/KDS flags, bulk assignment across items.

### E-Bills & Digital

Auto-send via SMS, WhatsApp, Email on settlement. QR code on receipt. 30-day secure link access.

### Online Ordering

White-label website, custom domain, mobile app (iOS/Android), OTP auth, promo codes, delivery zones.

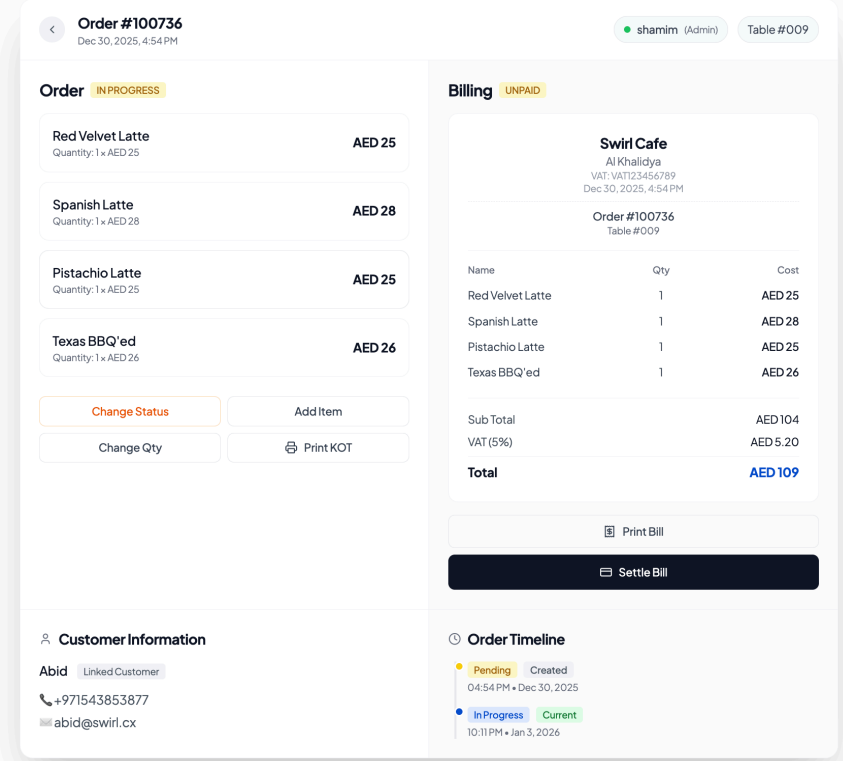
### Self-Service Kiosk

Touch-friendly UI, language selection, upsell prompts, payment terminal, loyalty scan, accessibility.

SWIRL DINE — POINT OF SALE

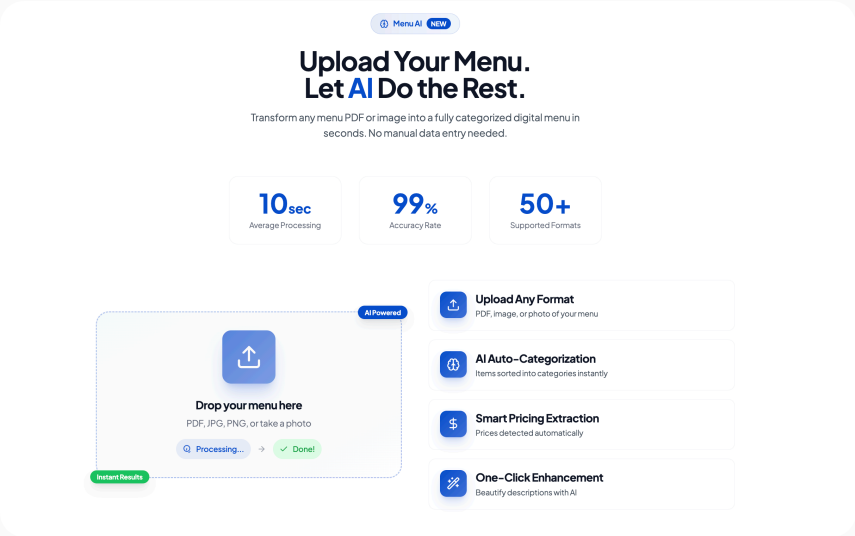
# More than a POS. Your restaurant's operating system.

Lightning-fast billing with deep integration into kitchen, inventory, accounting, and CRM. Every transaction is connected — no data gaps, no reconciliation headaches.

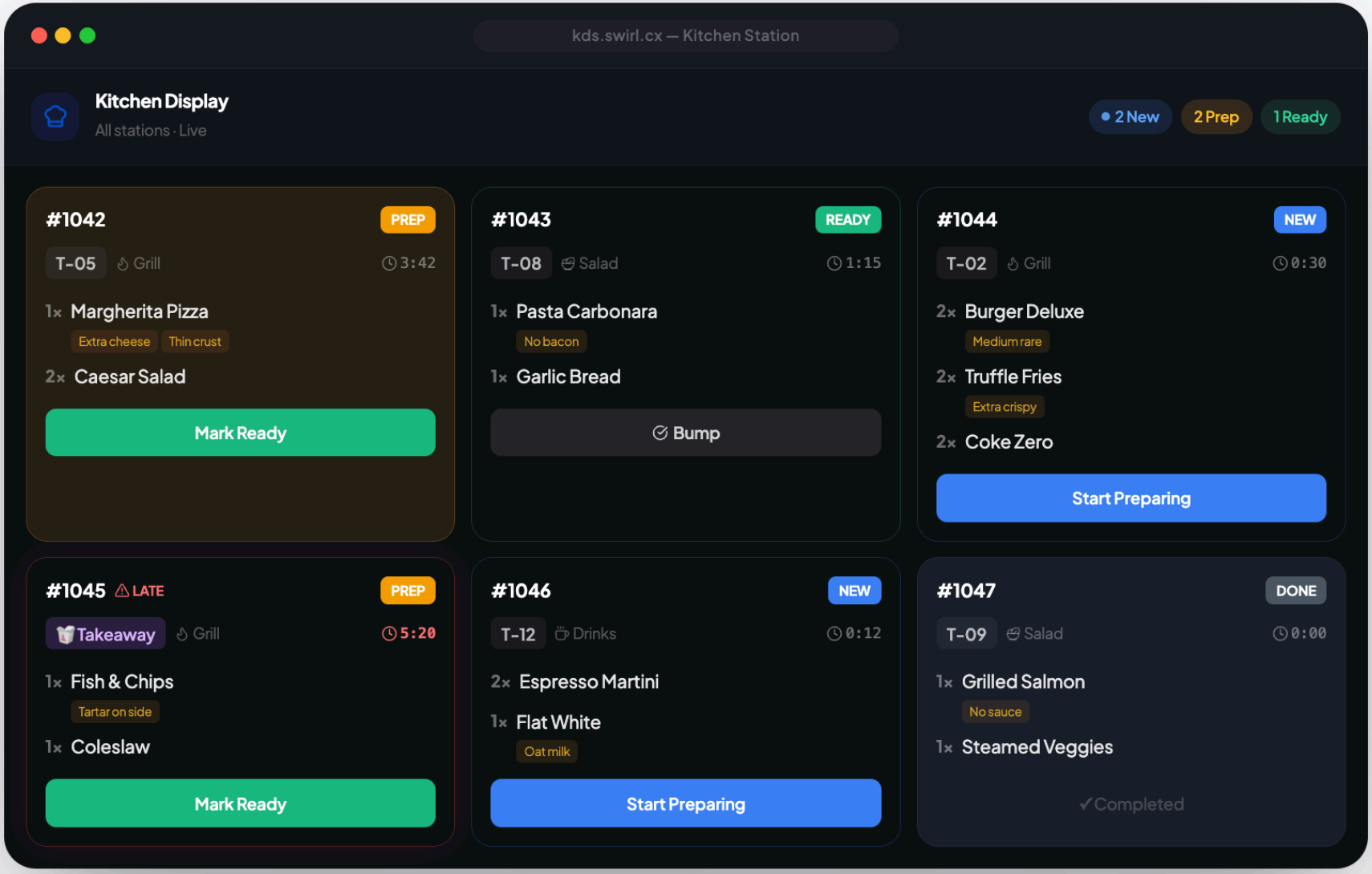


# Menus that optimize themselves.

AI-powered menu engineering that analyzes sales mix, food cost, and margin data to recommend pricing, placement, and promotions. Build menus once — Swirl keeps them profitable.



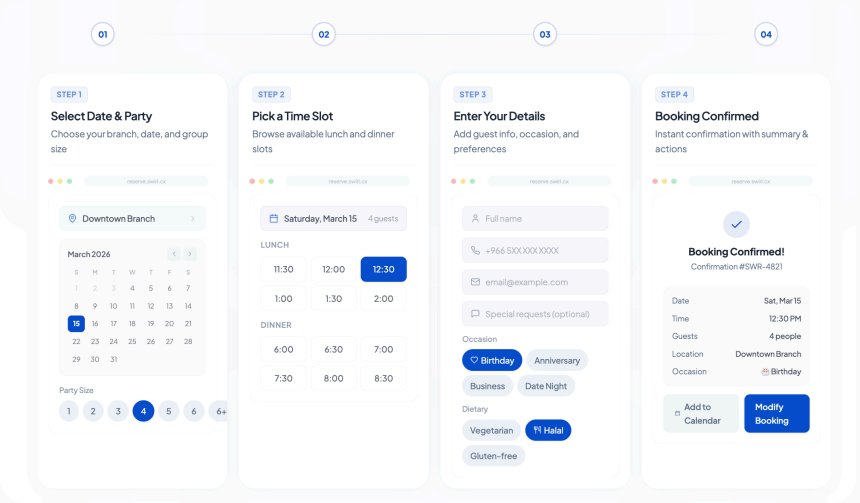
# From Order to Kitchen. In Real Time.



SWIRL DINE — GUEST EXPERIENCE

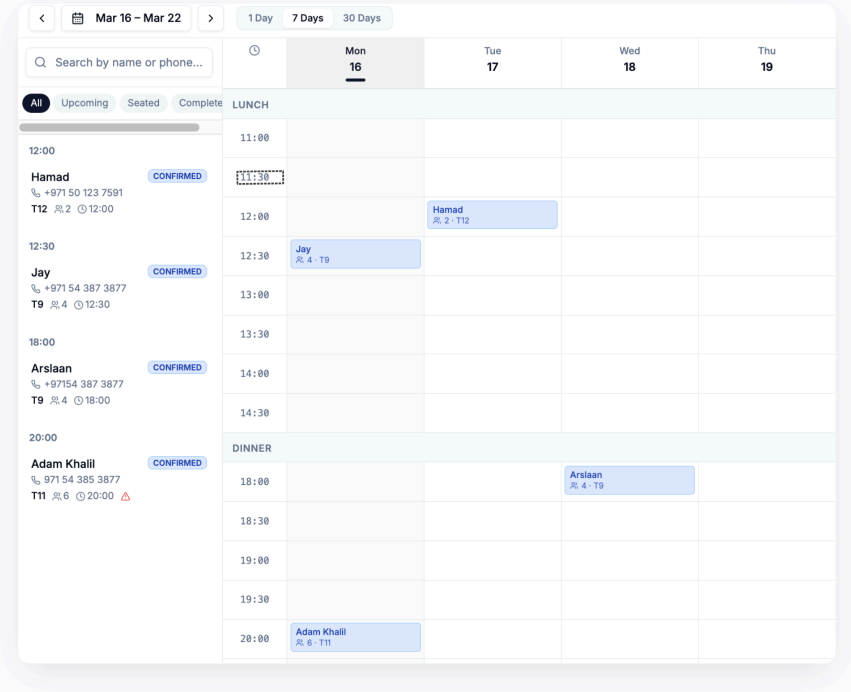
# Reservations your guests will remember.

A seamless booking experience for customers — from table selection to confirmation. Guest profiles, visit history, and preferences are captured automatically, turning every reservation into a personalized experience.



# Every table. Every time slot. One view.

A visual calendar that gives your host team full control — drag-and-drop bookings, real-time availability, walk-in management, and automatic waitlist handling. No more double-bookings or empty tables.




Timeline management | Service separation | Real-time availability | Table assignments | Booking blocks

# Less waste. More margin. Zero chaos.

Full control over the operational and financial backbone. From ingredient costing to vendor payments, every back-office function is automated, tracked, and optimized by AI.

### Inventory Management
Real-time stock tracking, par levels, batch management, automated alerts, recipe-based deduction on every sale.

### Procurement & Purchase Orders
Requisition-to-PO workflow, multi-level approvals, GRN with quality gates, three-way matching (PO-GRN-Invoice).

### Recipe & Food Costing
Ingredient-level recipes, unit conversions, cost per serving, theoretical vs actual variance analysis.

### Production & Transfers
Batch production runs, inter-store transfers with full workflow, multi-warehouse support, stock counting.

# Run everything. Manage nothing.

The intelligent layer that unifies finance, operations, supply chain, and HR into a single platform. Not a bolted-on ERP — a business operating system.

### Full Accounting

Chart of accounts, GL, journal entries, AR/AP, bank reconciliation, multi-currency, year-end close.

### Financial Statements

P&L, balance sheet, cash flow, trial balance. Group-level consolidation across entities.

### FTA Approved — ASP

Approved Service Provider (ASP) for UAE Federal Tax Authority. Full VAT compliance and e-invoicing readiness.

### Fixed Assets

Asset register, straight-line & WDV depreciation, maintenance tracking, disposal with gain/loss accounting.

### HR & Workforce

Attendance, scheduling, leave management, commission engine, tip distribution, payroll export (EFT).

### ZATCA E-Invoicing

Full Phase 2 compliance for Saudi Arabia. IRN generation, QR codes, real-time ZATCA portal submission.

# Built for scale. Designed for control.

### Multi-Entity Hierarchy

Group, country, division, company, branch. Intercompany transactions with elimination entries. Consolidated reporting.

### Multi-Brand Support

Operate multiple restaurant brands from one platform. Brand-specific menus, pricing, loyalty — shared back office.

### RBAC & Security

Granular role-based access per branch. Full audit trail on every transaction. Session management and password policies.

### Events & Banquets

Venue booking, function prospectus, resource allocation, advance tracking, post-event settlement and P&L.

### Training Mode

Sandboxed POS for staff training. No impact on live data. Performance tracking per trainee. Auto-purge after 30 days.

### Offline & Cloud

PWA with IndexedDB. Offline order entry, auto-sync, conflict resolution. Cloud-native with zero-downtime deploys.

### AI Intelligence

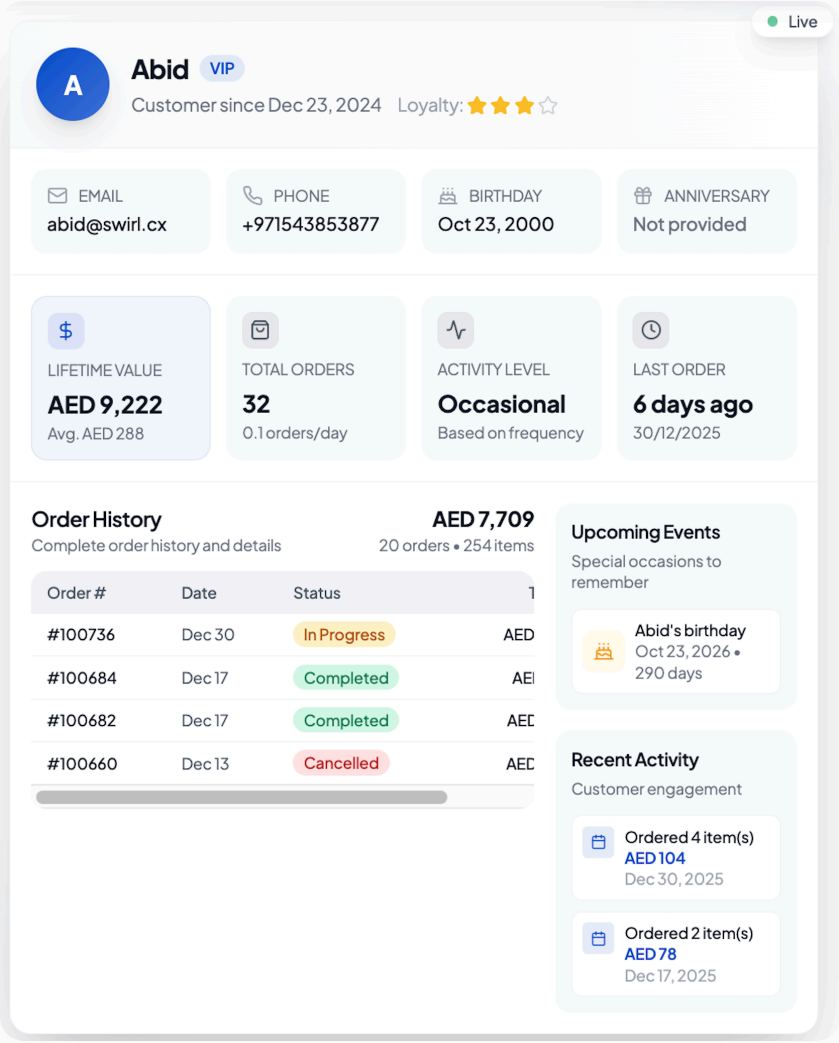
Demand forecasting, prep planning, inventory anomaly detection, recommendation engine, AI-powered command center.

### Multi-Language & RTL

Arabic with full RTL support. English. Translation management for menus, categories, and customer content.

# Know every guest. Before they walk in.

A 360° customer view — order history, spend patterns, visit frequency, feedback, and loyalty status in one place. Segment guests automatically, trigger personalized campaigns, and turn first-time visitors into lifetime regulars.



TECHNICAL FOUNDATION

# Modern stack. Enterprise grade.

### FRONTEND

Next.js 14 · React 18 · TypeScript · Tailwind CSS · Radix UI

### BACKEND & DATA

Node.js · Prisma ORM · PostgreSQL · Socket.io · REST API

### INFRASTRUCTURE

AWS S3 · Cloud Native · CDN · Auto-scaling · Zero-downtime Deploy

### INTEGRATIONS

UrbanPiper · Twilio · WhatsApp API · Sendgrid · Adyen · ZATCA

## Platform Numbers

| | |
|---|---|
| **438** | Database models — comprehensive domain coverage |
| **95+** | API endpoint categories — modular architecture |
| **100+** | Accounting models — full ERP capability |
| **5** | Order types with independent pricing engines |
| **3** | Loyalty program types (Points, Punch, Tiered) |
| **4+** | Payment terminal providers supported |
| **24/7** | Support with WhatsApp, email, and phone |

# Every dirham. Tracked and reconciled.

### Payment Reconciliation

5-tab dashboard: Transactions, Bank, PSP, Aggregator, Variances. Auto-confirm at 95%+ confidence match. Day-wise drill-down.

### Aggregator Settlement

SOA import, order matching, tiered commission verification, dispute management, and receivables tracking across all platforms.

### Three-Way Matching

Line-level PO-GRN-Invoice matching with configurable variance tolerance. Auto-flag at thresholds with approval workflow.

### Shift Reconciliation

Denomination-level cash counting, expected vs actual by payment method, auto-close scheduling, and HQ monitoring dashboard.

### Consumption Variance

Theoretical vs actual food cost by category, ingredient, and time period. Recipe-based deduction compared to actual inventory.

### Fraud Detection

Void frequency, discount patterns, refund anomalies, tip modifications, and reprint tracking with anomaly scoring and alerts.

# Turn guests into regulars. Automatically.

### RFM Segmentation

Automatic customer segmentation: Champions, Loyal, At Risk, Can't Lose, Lost, New. AI-driven scoring and segment rules.

### Multi-Channel Campaigns

Email (Sendgrid), SMS (Twilio), WhatsApp (Business API). Campaign builder with A/B testing, scheduling, and throttling.

### Trigger Automation

Birthday rewards, win-back after 30 days, welcome series, post-visit feedback, loyalty milestone celebrations — all automated.

### Cross-Brand Loyalty

Earn at Brand A, redeem at Brand B. Unified wallet with brand-specific earn/burn rates. Tier benefits shared or isolated.

### Guest Profiles

Preferences, allergies, spending patterns, visit frequency, favorite items. Available to every staff touchpoint in real-time.

### Feedback & NPS

Post-visit surveys, Net Promoter Score tracking, feedback analytics. Segment-based targeting for satisfaction monitoring.

# Go live fast. Stay supported.

Swirl's cloud-native architecture means rapid deployment without heavy infrastructure setup. Our team handles configuration, data migration, and staff training end-to-end.

- **24/7 Multi-Channel Support**
  WhatsApp, email, and phone. Dedicated account manager for enterprise clients.
- **Training Program**
  On-site and remote. Role-specific modules. Built-in Training Mode for hands-on practice.
- **SLA-Backed Response**
  Priority-based escalation (P1-P4). Automated alerting. Post-incident review for critical issues.
- **Continuous Updates**
  Quarterly feature releases. Zero-downtime deployments. Active 39-feature roadmap.

## Deployment Timeline

**Week 1-2** **Discovery & Configuration**

Requirements mapping, menu setup, user roles, payment terminal configuration, branch hierarchy setup.

**Week 3-4** **Data Migration & Integration**

Historical data import, aggregator onboarding, payment gateway setup, accounting chart migration.

**Week 5-6** **Training & Go-Live**

Staff training by role, UAT testing, pilot outlet launch, phased rollout across remaining outlets.

LET'S BUILD THE FUTURE TOGETHER

# Swirl

*Software that thinks.*

hello@swirl.cx

swirl.cx

Confidential. Prepared for enterprise evaluation. All product capabilities described reflect the current platform with active roadmap items noted where applicable.